An Intelligent Multi-Objective Framework for Undergraduate Project Team Formation and Project Feasibility Assessment

***Abstract*—Undergraduate project-team formation requires balancing technical suitability with student project preferences. This study proposes a project-aware multi objective allocation method using Heuristic Seeded NSGA-II. Student competency is represented through self reported technology ratings, while projects are modeled using explicit technical requirements. Technical requirement deficit and preference dissatisfaction are optimized separately to generate nondominated alternatives rather than a single weighted solution. Surplus approved projects are supported through bounded project-subset selection. Evaluation included exact Pareto verification, baseline comparison, scalability testing, and a real-response case study. For a controlled 12 student, three-project instance, the method recovered the exact Pareto front in the verified conflict scenarios and achieved an overall mean hypervolume of 0.9479 across the controlled evaluation. A synthetic 1,000 student, 250 project test produced 25 nondominated alternatives in approximately 30.15 s. In a 60 student real response case study, the project technical requirements were reviewed and approved before evaluation; the method achieved 100% modeled technical coverage and 95% preference satisfaction.**

***Index Terms*—Team Formation, Multi Objective Optimization, NSGA-II, Project Allocation, Project Feasibility**

I. Introduction

The formation of undergraduate project teams requires more than distributing students into groups of equal size. Academic staff must consider whether each team collectively possesses the technical competencies needed by its assigned project while also accounting for students' preferred topics. These criteria can conflict: a technically strong allocation may reduce preference satisfaction, whereas a preference oriented allocation may leave project requirements insufficiently covered. The problem also grows combinatorially; assigning only 12 students to three unique projects in teams of four already produces 34,650 feasible complete allocations.

Existing educational team formation research has considered cohesion, personality, behavioral characteristics, and learning data [1]–[4]. Software engineering staffing research has additionally considered technical competencies, project requirements, and multi-objective optimization [5]–[8]. However, limited attention has been given to undergraduate project allocation where *project technical suitability* and *student project preference* are retained as separate competing objectives and exposed as multiple nondominated alternatives instead of being collapsed into a fixed weighted score.

To address this gap, this study proposes a project-aware team formation and project allocation method using Heuristic-Seeded NSGA-II. Student capability is represented through self reported ordinal competency ratings, while project requirements specify a technology, minimum competency threshold, and required number of qualified members. Two objectives are minimized separately: technical requirement deficit (f_1) and preference dissatisfaction (f_2). The method also supports situations in which approved projects outnumber required teams through a bounded outer project subset procedure.

The contributions are: (1) an interpretable representation linking competency indicators to project technical requirements; (2) a two objective formulation preserving technical and preference criteria separately; (3) a heuristic-seeded NSGA-II allocation procedure extended for surplus approved projects; and (4) evaluation combining exact Pareto verification, baseline and statistical comparison, scalability testing, and a 60 student real-response case study using reviewed and approved project technical requirements.

The study addresses three research questions. **RQ1** asks how student competencies can be matched to explicit project technical requirements. **RQ2** examines how effectively the formulation exposes technical-versus-preference trade-offs. **RQ3** evaluates how V3 compares with greedy, random, weighted, and unseeded baseline approaches.

II. Related Work

Educational team formation studies have examined cohesion, student characteristics, and algorithmic grouping. Shaikh identified 128 criteria for cohesive software-engineering capstone teams [1]. Løvold *et al.* compared student-selected and instructor-assigned teams in a large software-engineering course [2]. Sanchez-Anguix *et al.* compared classroom grouping based on MBTI- and Belbin-oriented heuristics [3], while Liang *et al.* evaluated a mixed genetic algorithm in an authentic classroom using learning-log data [4]. These works automate educational grouping but do not center the allocation on explicit technical requirements of alternative projects.

Requirement aware personnel allocation is more common in software engineering. André *et al.* modeled competencies, roles, availability, and project constraints [5]. Paredes-Valverde *et al.* used an ontology based approach for project-to-personnel matching [6]. Stylianou and Andreou combined technical and personality related criteria in multi objective staffing [7], whereas Arunachalam *et al.* used NSGA-II for agile team selection with cost and productivity objectives [8].

Further optimization studies include Moreno *et al.*, who used a genetic algorithm with multiple student characteristics [9], and Casotti and Krohling, who compared Pareto based evolutionary algorithms for a bi-objective team-formation problem using hypervolume [10]. The present work instead targets explicit undergraduate project requirements and ranked project preferences.

TABLE I
POSITIONING RELATIVE TO REPRESENTATIVE PRIOR WORK

Study	Context	Primary focus	Approach
Shaikh [1]	Education	Team cohesion criteria	Criteria-based
Liang et al. [4]	Education	Learning activity data	Mixed GA
André et al. [5]	Professional	Competencies, roles, constraints	Formal model
Paredes-Valverde et al. [6]	Professional	Competency-requirement matching	Ontology-based
Stylianou and Andreou [7]	Professional	Technical skills and personality	Multi-objective GA
Proposed	**Undergraduate**	**Technical requirements + project preference**	**Multi-objective NSGA-II**

Table I highlights the complementary nature of prior work: educational approaches primarily study student/team attributes, while professional staffing studies provide stronger competency and requirement-aware models. In the reviewed literature, limited attention is given to undergraduate allocation where project technical suitability and project preference remain separate competing objectives and are exposed as Pareto alternatives for academic staff.

III. METHODOLOGY

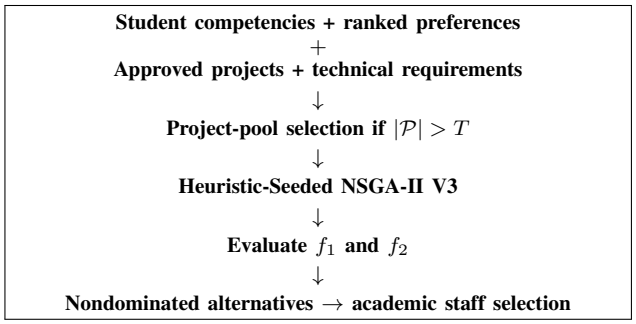


Fig. 1. Research workflow of the proposed team formation method.

A. Problem Formulation and Input Representation

Let $\mathcal{S}$ denote the set of N students and $\mathcal{P}$ the approved projects. Each student has self-reported competency ratings $s_{ik} \in \{1, \ldots, 5\}$ across 16 technologies and a ranked project-preference list. Each project p has requirements $\mathcal{R}_p$; requirement k specifies a minimum competency L_{pk} and required number of qualified members M_{pk}.

For the 60 student real response case study, the overall scenario and all 15 project technical-requirement configurations including technologies, minimum competency levels, and required member counts were reviewed and approved before evaluation.

Given target team size b, the number of teams is

$$T = \left\lceil \frac{N}{b} \right\rceil. \tag{1}$$

A smaller remainder team is permitted when N is not divisible by b. Let x_{it} indicate student-to-team assignment, y_{tp} team-to-project assignment, and $\mathcal{P}_s \subseteq \mathcal{P}$ the selected projects. Feasibility requires

$$\begin{aligned} &\sum_{t=1}^{T} x_{it} = 1 && \forall i \in \mathcal{S}, \\ &\sum_{p \in \mathcal{P}_s} y_{tp} = 1 && \forall t, \\ &\sum_{t=1}^{T} y_{tp} \le 1 && \forall p \in \mathcal{P}_s, \\ &|\mathcal{P}_s| = T, \quad \mathcal{P}_s \subseteq \mathcal{P}, \quad T \le |\mathcal{P}|. \end{aligned} \tag{2}$$

Thus, every student is assigned exactly once, each team receives one unique selected project, surplus approved projects may remain unassigned, and $T > |\mathcal{P}|$ is infeasible.

B. Technical Requirement Deficit

A student qualifies for requirement (p, k) when $s_{ik} \ge L_{pk}$. For team G_p assigned to project p, define

$$q_{ipk} = \begin{cases} 1, & s_{ik} \ge L_{pk}, \\ 0, & \text{otherwise}, \end{cases} \qquad N_{pk} = \sum_{i \in G_p} q_{ipk}. \tag{3}$$

Requirement coverage, project coverage, and project deficit are

$$\begin{aligned} &C_{pk} = \min\left(1, \frac{N_{pk}}{M_{pk}}\right), \quad C_p = \frac{1}{|\mathcal{R}_p|} \sum_{k \in \mathcal{R}_p} C_{pk}, \\ &D_p = 1 - C_p. \end{aligned} \tag{4}$$

The first objective is the mean deficit across selected projects:

$$f_1 = \frac{1}{|\mathcal{P}_s|} \sum_{p \in \mathcal{P}_s} D_p. \tag{5}$$

The optimizer minimizes f_1; $f_1 = 0$ denotes complete coverage of the modeled requirements. The threshold-based formulation avoids treating ordinal self reported competency values as exact linear measurements and should not be interpreted as predicting project success or grades.

C. Preference Dissatisfaction

Let $r_i(p)$ be the rank assigned by student i to project p. When all participating projects are ranked,

$$d_i(p) = \frac{r_i(p) - 1}{|\mathcal{P}_s| - 1}. \tag{6}$$

For a partial Top-K list,

$$d_i(p) = \begin{cases} (r_i(p) - 1)/K, & p \text{ is ranked}, \\ 1, & p \text{ is unranked}. \end{cases} \tag{7}$$

Thus, for Top 3 preferences with more than three participating projects, the first, second, and third choices receive penalties 0, $1/3$, and $2/3$. If $a(i)$ denotes the assigned project,

$$f_2 = \frac{1}{N} \sum_{i \in \mathcal{S}} d_i(a(i)), \qquad \mathbf{F} = (f_1, f_2). \tag{8}$$

Both objectives are minimized independently rather than combined using a fixed weighted score, allowing the optimizer to expose technical-versus-preference trade-offs.

D. Heuristic Seeded NSGA-II

The search is based on NSGA-II, an established elitist multi objective evolutionary algorithm using nondominated sorting and crowding distance [11]. The contribution is therefore the project aware formulation, objective design, heuristic initialization, and decision support workflow rather than NSGA-II itself.

For a fixed selected project set, one chromosome is a permutation of all students partitioned into team slots. V3 uses a population of 120 and 150 generations. Order crossover [12] preserves the permutation structure, while mutation performs a cross team student swap.

Rather than initializing entirely at random, V3 combines a technical greedy seed, preference greedy seed, nearby swap variants, and random feasible individuals. Heuristic solutions act only as initialization anchors; subsequent selection is governed by f_1, f_2, nondominated sorting, and crowding distance. The result is a set of nondominated alternatives from which academic staff select the final allocation rather than a single automatically weighted solution.

E. Approved Project Pool Selection

When $|\mathcal{P}| = T$, all approved projects are passed directly to V3. When $|\mathcal{P}| > T$, project selection is handled through a bounded outer candidate-subset procedure without modifying the evaluated V3 student permutation representation or evolutionary operators.

Let $B = 8$ be the default maximum number of candidate subsets. If

$$\binom{|\mathcal{P}|}{T} \leq B, \tag{9}$$

all subsets are evaluated; otherwise, candidates are constructed using preference oriented and technical potential anchors, one-project neighborhood variants, and reproducible random subsets. V3 is run for each subset, after which all returned solutions are pooled and globally filtered by Pareto dominance on (f_1, f_2). Technical deficit is evaluated only for selected projects, while preference dissatisfaction is evaluated for all students.

For large surplus project cases, the outputs are therefore described as *discovered nondominated alternatives* or an *approximated nondominated set*, not a guaranteed complete global Pareto front.

F. Search Procedure and Human Decision Support

For a fixed project set, chromosomes encode student permutations that are partitioned into team slots. Each candidate is decoded, evaluated using (f_1, f_2), and ranked by NSGA-II nondominated sorting and crowding distance. Heuristic seeds provide promising starting regions but do not alter the fitness definition or receive preference after initialization. For surplus project cases, at most B candidate project subsets are optimized independently with the same V3 configuration; their returned solutions are then pooled and filtered again by Pareto dominance.

The final output is a decision-support set rather than an automatic academic decision. Staff can compare team membership, assigned project, technical coverage, preference evidence, and objective values before selecting an allocation. Technical Coverage Inspection is an explanatory post-optimization feature, while supervisor allocation is a separate downstream process based on capacity, expertise, research interest, and workload. Neither process modifies f_1 or f_2. This separation keeps the evaluated research formulation distinct from later operational decisions.

IV. Evaluation and Results

The method was evaluated through exhaustive verification, baseline comparison, synthetic scalability testing, surplus project constraint verification, and a real response case study. Controlled experiments evaluated correctness and objective space quality, while the real response study examined applicability with naturally occurring competency and preference data.

A. Experimental Design

Synthetic cohorts of 20, 40, 60, and 100 students were generated under low, medium, and high technical versus preference conflict. Ten dataset seeds were evaluated for every cohort size/conflict combination, producing 120 controlled datasets. Stochastic methods used three optimizer seeds. V3 used population 120 and 150 generations.

Comparators were Technical Greedy, Preference Greedy, Random Search, a Weighted Single Objective genetic algorithm, and Unseeded NSGA-II. Random Search and the weighted GA received comparable evaluation budgets. Hypervolume [10] was the principal multi objective metric, with higher values indicating better objective space coverage. The complete controlled evaluation comprised 1,680 experiment records.

To support reproducibility, controlled datasets were generated from recorded dataset seeds and stochastic methods were repeated with three optimizer seeds under the same population and generation settings. The real response sample used fixed selection seed 20260829, while the 12 student verification cases were independently exhaustively enumerated rather than relying only on stochastic search.

TABLE II
CONTRIBUTION-TO-EVALUATION TRACEABILITY

ID	Contribution / question	Primary evidence
C1/RQ1	Competency-to-requirement representation	Objective formulation + validated 60-student case
C2/RQ2	Separate f_1 and f_2 objectives	Exact Pareto fronts + conflict-level front sizes
C3/RQ3	Heuristic-seeded search	Baselines + unseeded ablation-style comparison + statistics
C4	Practical allocation at scale / surplus projects	1,000/250 scalability + project-pool integrity checks

B. Evaluation Rationale

The evaluation was intentionally layered because different experiments support different claims. Exhaustive search provides correctness evidence when the true Pareto front is computationally obtainable. Controlled synthetic experiments support baseline comparison under repeatable conflict levels and cohort sizes. The 1,000 student experiment evaluates computational scalability, while the real response case study examines whether the same formulation can process naturally occurring linked competency and preference responses. These results are interpreted separately rather than combined into a single claim of general effectiveness.

Hypervolume was selected as the main comparative metric because the method returns a set of nondominated alternatives rather than one solution. Runtime, objective endpoints, nondominated-set size, and allocation integrity complement hypervolume. Integrity checks verify complete and unique student assignment, valid team sizes, and unique selected projects.

C. Exact Pareto Verification

For 12 students, three projects, and teams of four, all 34,650 feasible complete allocations were exhaustively evaluated. The exact Pareto points were

$$\begin{aligned}\mathcal{F}_M^* &= \{(0, 0.20833), (0.05556, 0.125), (0.22222, 0.08333)\},\\ \mathcal{F}_H^* &= \{(0, 0.375), (0.05556, 0.29167), (0.22222, 0.20833),\\ &\qquad (0.38889, 0.16667)\}.\end{aligned} \tag{10}$$

V3 recovered the complete exact Pareto front in all final repeated medium and high conflict verification runs, confirming recovery of the known nondominated sets when exhaustive verification was possible.

D. Baseline Comparison and Seeding Ablation

Table III summarizes overall mean hypervolume. V3 achieved the highest value, while Preference Greedy was the strongest deterministic baseline.

The comparison with Unseeded NSGA-II provides an ablation-style test of the principal initialization contribution: heuristic initialization is removed while the underlying NSGA-II framework is retained. V3's mean hypervolume of 0.9479 exceeded the unseeded value of 0.6532 by approximately 45.1%, indicating a material benefit from heuristic seeding.

TABLE III
OVERALL MEAN HYPERVOLUME BY METHOD

Method	Mean Hypervolume
Heuristic-Seeded NSGA-II V3	**0.9479**
Preference Greedy	0.7793
Unseeded NSGA-II	0.6532
Weighted Single-Objective GA	0.6024
Technical Greedy	0.3721
Random Search	0.3632

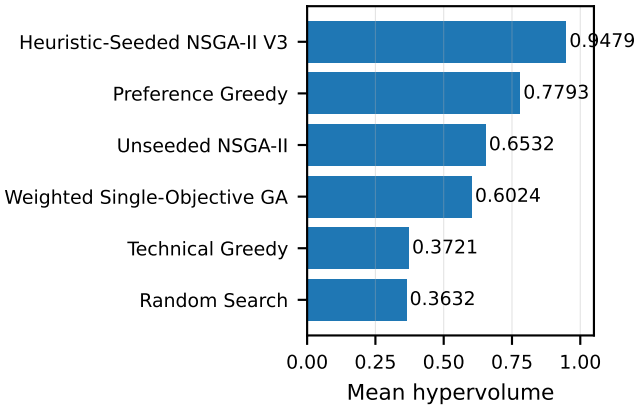


Fig. 2. Overall mean hypervolume across the evaluated allocation methods.

This is a framework-level seeding ablation rather than a separate ablation of each individual seed source.

For the four stochastic methods, a Friedman test on dataset-level median hypervolumes found a significant difference ($\chi^2(3) = 326.08$, $p < 0.001$, Kendall's $W = 0.906$). Holm corrected one sided paired Wilcoxon tests showed significantly higher hypervolume for V3 than Unseeded NSGA-II, Weighted Single Objective GA, and Random Search ($p < 0.001$ for all three comparisons). Mean ranks were 1.1417, 2.0375, 2.8208, and 4.0000, respectively.

Mean V3 nondominated set sizes were 1.00, 13.21, and 28.78 for low, medium, and high conflict, respectively, showing that richer alternative sets emerge as objective conflict increases.

The statistical findings strengthen the baseline comparison but are kept within the scope of the controlled experiments. The Friedman result shows that the stochastic methods did not exhibit equivalent hypervolume performance, while Kendall's $W = 0.906$ indicates a strong method-ordering effect in this evaluation. Holm-corrected pairwise tests further support V3's advantage over the stochastic comparators. These results do not establish universal superiority beyond the evaluated problem settings.

E. Scalability Evaluation

A synthetic scalability test used 1,000 students, 250 projects, and 250 teams of four. Approximately 16,744 objective evaluations produced 25 discovered nondominated alternatives; all passed integrity checks for student completeness and uniqueness, team size, and unique project assignment.

The optimized NSGA-II stage required approximately 28.9 s and the full team-formation pipeline 30.15 s, compared with 1,336.7 s for the earlier implementation ($46.3\times$ speedup). This implementation optimization did not change the objectives or evolutionary search configuration. Because the dataset was synthetic, the result is evidence of computational scalability rather than institutional deployment performance.

F. Surplus Project Constraint Verification

Implementation-level regression tests verified the project pool constraints independently of optimization-quality claims. With 60 students and 15 approved projects, target team size 10 produced six teams and therefore allowed nine projects to remain unassigned; target size 7 produced nine teams including a remainder team of four; and target size 3 correctly failed because 20 unique project teams would be required. The exact-count case of team size 4 retained the original 15 project path. These checks verify structural correctness of the surplus project extension rather than its global optimization quality.

G. Real-Response Case Study

A single combined anonymous survey collected 16 self reported technology competency ratings and three ranked project preferences from each respondent, preserving competency-preference correspondence. Sixty six complete responses were received and were structurally valid. Sixty were selected using simple random sampling without replacement with fixed seed 20260829, with six retained as reserves. The case study used 15 project options and teams of four.

The complete scenario and all 15 projects' technical requirements were reviewed and approved before evaluation, covering technologies, minimum competency thresholds, and required qualified-member counts.

The optimizer produced

$$(f_1, f_2) = (0, 0.05), \tag{11}$$

corresponding to 100% modeled technical requirement coverage and 95% preference satisfaction. Independent calculation also established $f_2 = 0.05$ as the minimum achievable preference dissatisfaction under the capacity constraints. Thus, one feasible allocation simultaneously attained both objective minima.

The single nondominated objective point therefore reflects an absence of necessary objective conflict in this dataset rather than optimization failure; the deliberately constructed medium and high conflict cases produced multiple Pareto alternatives.

V. Discussion and Conclusion

A. Discussion

The results demonstrate that maintaining Technical Requirement Deficit and Preference Dissatisfaction as separate objectives provides interpretable decision support for undergraduate team project allocation. V3 recovered exact Pareto fronts when exhaustive verification was possible and achieved the strongest overall hypervolume among the compared approaches. The growth in nondominated-set size from low to high conflict further suggests that the formulation responds to genuine technical versus preference conflict rather than artificially generating alternatives.

The 60 student case study provides complementary empirical evidence. Linked competency and preference responses were used, and the technical requirements for all 15 projects were reviewed and approved before evaluation. The allocation $(f_1, f_2) = (0, 0.05)$ simultaneously reached complete modeled technical coverage and the independently established preference minimum. The seeded versus unseeded comparison additionally indicates that the heuristic initialization contributes materially to objective space coverage.

The surplus project extension increases practical applicability by allowing approved projects to remain unused when project supply exceeds team demand. The system remains staff oriented decision support: academic staff retain final allocation authority, while Technical Coverage Inspection and downstream supervisor allocation remain separate from the two optimization objectives.

B. Practical Implications and Decision Support Use

The proposed method supports rather than replaces academic decision making. Instead of producing one mandatory allocation, it presents nondominated alternatives that expose the trade off between technical requirement coverage and student preference. Because coverage is traced to explicit technologies, competency thresholds, required member counts, and qualifying students, staff can inspect why an allocation is technically strong or where gaps remain before confirming it.

The surplus project mechanism also reflects the practical case where approved topics outnumber student groups by allowing some projects to remain unassigned. For large surplus cases, the bounded search means the returned alternatives are discovered within the evaluated search procedure rather than guaranteed to form the complete global Pareto set. The selected allocation can then support downstream supervisor assignment and project planning without changing the two evaluated optimization objectives.

C. Limitations

The principal limitations are that student competencies are self reported ordinal indicators; the real response evaluation uses one 60 student cohort; and the bounded surplus-project search does not exhaustively evaluate all project subsets in large cases. The reported 1,000 student scalability experiment used equal project and team counts, so additional large scale empirical evaluation of surplus-project selection is warranted. Furthermore, the current ablation isolates the presence or absence of heuristic seeding as a whole rather than the individual contribution of the technical seed, preference seed, and local swap variants. These limitations restrict claims of institution wide or globally optimal performance.

D. Threats to Validity

Construct validity: Technical Requirement Deficit measures coverage of modeled technology requirements, not

project success, grades, team cohesion, or productivity. Preference Dissatisfaction represents ranked topic preference rather than overall student satisfaction. Competencies are self-reported ordinal indicators; consequently, threshold qualification is used instead of treating score differences as precise interval measurements.

Internal and conclusion validity: Controlled datasets use recorded seeds, stochastic methods are repeated, and exact 12-student enumeration provides an independent reference for Pareto recovery. Comparable evaluation budgets were used for stochastic baselines. However, three optimizer seeds are still a limited sample, and the current ablation removes heuristic seeding as a whole rather than isolating each seed source. Statistical tests therefore support the evaluated configuration but not every possible NSGA-II setting.

External validity: The 1,000 student experiment is synthetic and demonstrates computational scalability rather than institutional effectiveness. The real response study uses one 60-student cohort and one 15 project case. Review of the project requirements strengthens the case study requirement specification but is not institution-wide validation. Moreover, large surplus project cases use bounded subset search rather than exhaustive enumeration. Broader claims therefore require replication across institutions, cohorts, and project pools.

E. Conclusion and Future Work

This study proposes a project aware Heuristic-Seeded NSGA-II approach that combines explicit technical requirements with student project preferences while preserving both as separate objectives. Evaluation demonstrated exact front recovery in controlled cases, superior overall hypervolume against the evaluated baselines, scalability to 1,000 students and 250 projects in approximately 30.15 s, and applicability to a 60 student case study using reviewed project requirements.

Future work should examine larger multi institution cohorts, competency evidence beyond self report, component-level seeding ablations, longitudinal project outcomes, and more extensive project-subset and fairness-aware optimization. Institutional studies should also examine staff acceptance, decision time, and perceived transparency.

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